LLM-Based Anomaly Detection in System Logs

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Abstract

In this project, LLMs will be applied in a software application to identify abnormal patterns in network and system logs. The practical application that was deveoped to test LLM anomaly detection is a web-based system designed using the MongoDB, Express, React, and Node (MERN) stack. The framework that will be used to scan the network logs is called LangChain combined with Claude. The topic number in this syllabus is: 14. LLM-Based Anomaly Detection in System Logs. So far the application has been built with Backend Logging using the Javascript logging Library Winston which so far is able to log backend API requests and responses. LangChain framework has been implemented to some extent and is able to scan logs. However more fine-tuning needs to be done as it is not consistent and there are still bugs and issues with the Claude API.

CCS Concepts

• **Computing methodologies** → **Natural language processing**; *Neural networks*; • **Security and privacy** → **Intrusion/anomaly detection and malware mitigation**; • **Software and its engineering** → System administration.

Keywords

large language models, anomaly detection, log analysis, system monitoring, LangChain, natural language processing, security analysis, network logs

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1 Code Availability

The source code for the implementation of this web application can be found at:

https://github.com/lamcnguyen89/CAP_6640_AI_Logging

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2 Introduction and Problem Statement

2.1 Motivation

The motivation for this project is to explore how LLMs can be used to identify problems in system and network logs, which can be a valuable tool for system administrators and security professionals.

2.2 Significance

The significance of this project is that it can help to improve the efficiency and effectiveness of log analysis, which can lead to faster identification of issues and improved security.

2.3 Objectives

The objectives of this project are to develop a web application that can scan network logs using LangChain and Claude, and to evaluate the performance of the application in identifying anomalies in the logs.

3 Related Work

Recent advances in applying large language models to log analysis have shown promising results.

Egersdoerfer et al. [1] began using LLMs such as ChatGPT for log analysis since they are able to retain context and formulate insightful responses over conversations. These LLMs are able to conduct few-shot and in-context learning with reasoning ability. Previously there were several problems with pre-existing techniques. The first issue was that there was a heavy reliance on expert-labeled logs to discern anomalous patterns. This manual labelling slowed down training. The second issues was that they relied on numeric model prediction based on numeric vector input which is not human readable. This lack of human readability rules out targeted error correction by humans.

Qi et al. [4] introduced LogGPT. This paper explored the use of ChatGPT for log-based anomaly detection. It demonstrated that ChatGPT could be effectively used to identify anomalies in system logs, showing promising results in terms of accuracy and efficiency compared to traditional methods.

Guan et al. [2] introduced LogLLM.Traditional deep-learning methods often struggle to capture semantic information in log messages which are typically organized in natural language. LogLLM employed BERT for extracting semantic vectors from log messages, and uses Llama for classifying log sequences. It also uses a projector to align the vector spaces of BERT and Llama to ensure consistency in understanding log semantics. LogLLM doesn't require log parsers to extraxt templates from logs.

He et al. [3] introduced LLMeLog. This paper introduced a system intended to solve two problems. The first problem is that semantic noise is buried in log events which lead to inevitable gaps between the semantics from the logs and their actual intended meaning.

The second problem was that there exists a gap between general semantic embeddings and the specific embedding requirement for anomaly detection tasks. LLMeLog uses LLMs to enrich the content of log events with in-context learning techniques. These enriched logs are then used to fine-tune a pre-trained BERT model. Finally, it trains a transformer-based anomaly detection model with the event representationas produced by this fine-tuned BERT model. F1 scorese exceeded 99 percent and even with using only 10 percent of the labeled data, was still able to achieve an F1 score of 90 percent.

4 Methodology

In order to demonstrate how LLMs can be used to identify problems in system and network logs, a web application will be developed using the MERN stack. The application will then utilize some LLM log frameworks that were developed for LLM log analysis to scan a repository of logs and identify anomalies. An attempt will be made to balance performance with cost efficiency. The performance of the application will be evaluated using standard evaluation metrics for anomaly detection which is to identify errors and odd behaviors like excessive logins and failed login attempts.

4.1 Overview of Web Application

Before going into the logging techniques, a discussion will be made about the type of example application that will be built to generate the logs. This is important because the architecture of the application will determine the approach used to create the LLM log-analysis system.

The application that will be used is a web based service built using ther MERN stack. The MERN stack for web development is a popular choice for building web based applications. MERN stands for MongoDB, Express, React and Node. These are various frameworks that handle different parts of the pipeline. Before going into detail about What each of these frameworks do, you need to understand the overall architecture of a web based application. A web based application consists of a front end User Interface. This front end is where a user will interact directly with the application.

It consists of a front-end user interface that the users will directly interact with through a web browser. In this front end section of the application, the user will essentially be able to Create, Read, Update and Delete data. The front end of the software can be compared to the front of a grocery store where customers can look at products, add them to their carts, ask questions to associates and purchase products.

Beneath the front end is the middle layer of the application. This middle layer is the connection between the front end and the back end (Server). In this middle layer, API requests that perform translate the user's interactions into API requests that are sent to the backend server. It is like a middle translation layer that translate the front end user interactions into commands that the backend code and architecture can understand. Using the gorcery store analogy, the middle layer is like the associates that take customers requests for products, complaints and questions and translate them into requests for managers or the warehouse. These middle level associates understand a little of the both customer facing roles and the warehouse roles and are able to translate between the two.

Beneath the middle layer of the application is the backend portion of the code. The backend portion of the code using the analogy of the grocery store is like the warehouse where products are stored and can be retrieved and sent to the front of the store. In terms of software, this is where user data is stored and can be retrieved. It's where things like website articles, banking information, or whatever is stored inside a database. The backend code is also where the API requests from the middle layer are processed and executed.

So with this 3 part architecture of the web application in mind, we can understand how the different parts of the MERN stack fit into this architecture.

4.2 Overview of the MERN Stack

MERN stands for MongoDB, Express, React and Node. These are various frameworks that handle different parts of the pipeline.

First let us discuss the front end framework which is React. React is a javascript library used for building the User Interface of web applications. It is used in applications like Facebook, Instagram and even Amazon. The main takeway about why React is used to build front end applications is because it is designed to build very scalable web applications. React is built in such a way that 1 web page for a user can be automatically generated, or 1 billion web pages for 1 billion users can be automatically generated without having hand code each individual web page. The react code is used to produce the front end user interface that users will interact with.

The next technology out of the MERN stack to discuss is Express. Express is a web application framework that acts as the middle layer of the application. It allows the front end react code to communicate with the back end server code that stores and handles the database as well as other functions that the end user won't necessarily see. Express is used to create something called API endpoints that can speak to the database. API endpoints are like prebuilt messages that allow a specific set of actions to interact with the database. They translate the user's interactions into code that manipulates the database.

The next technology to discuss is MongoDB. MongoDB is a NoSQL database that is used to store data for the application. It is a document-oriented database that stores data in JSON-like format. It is used to store user data, website articles, and other information that the application needs to function. MongoDB can store music, videos, images and other types of data. It is used to store the data that the application needs to function.

The final technology in the MERN stack is Node. Node is a javascript runtime that allows developers to write front end code and back end code in the same Javascript programming language. Without it, developers would have to write front end code in one programming language, and the back end code in another programming language. This streamlines the process of development

In summary, the MERN stack is a set of technologies that are used to build functional web applications that deal with creating, reading, updating and deleting data. But what we now have to consider is the process of logging and log analysis when using these web applications.

4.3 Logging and LLM Log Analysis

The technique for working with Logging involves 4 steps.

The first step is to collect and possibly parse the logs. The way the application is designed, all the logs will be generated and downloaded into a single location. The logs will then be parsed and preprocessed to ensure that they are in a format that can be easily analyzed by the LLM.

The second step is to perform Vector Search and Contextual Retrieval. This step provides a baseline of what normal operations shown in the logs will look like. Retrieval augmented generation (RAG) is a technique that can be used to improve the performance of LLMs by providing them with relevant context from the logs. This can be done using vector search techniques to retrieve relevant log entries based on their semantic similarity to the input query. RAG will be used to feed the LLM.The LLM that will be used is Claude.

The third step is to perform Anomaly Detection. This step involves using the LLM to analyze the logs and identify any anomalies or patterns that may indicate a problem. The LLM can be trained to recognize normal patterns in the logs and flag any deviations from those patterns as potential anomalies.

The fourth step is to analyze the root cause of all the errors from the series of logs and explain in human readable language what the problem is and how to fix it.

4.4 Evaluation Metrics

The performance of the application will be evaluated using standard evaluation metrics for anomaly detection, such as precision, recall, and F1 score. These metrics will be calculated based on the number of true positives, false positives, and false negatives identified by the application.

For review, precision is the ratio of true positives to the total number of positive predictions made by the model. Recall is the ratio of true positives to the total number of actual positive cases in the dataset. The F1 score is the harmonic mean of precision and recall, providing a single metric that balances both aspects of performance.

5 Evaluation and Results

So far the application has been built with Backend Logging using the Javascript logging Library Winston which so far is able to log backend API requests and responses. LangChain framework has been implemented to some extent and is able to scan logs. However more fine-tuning needs to be done as it is not consistent and there are still bugs and issues with thee Claude API.

You can see in Figure 1 the user interface of the application. The application is essentially a web application that keeps track of experiments and experiment data. It is a standard web application with Create, Read, Update and Delete functionality. Data is stored in a database and logs are generated everytime there is communication between the front end of the code which is the website and the backend server.

In Figure 2 you can see that the logs are generated and saved in a single location in a database. The logs are generated using the Winston logging library and are in JSON format. The logs include information about the API requests and responses, as well as any errors that may occur. They are also stored in a separate CSV file that is used for the LangChain framework to scan and analyze the logs. This might not be the most efficient, but for the purposes of this project where we are focused more on implementing LLM Log Analysis, this is sufficient.

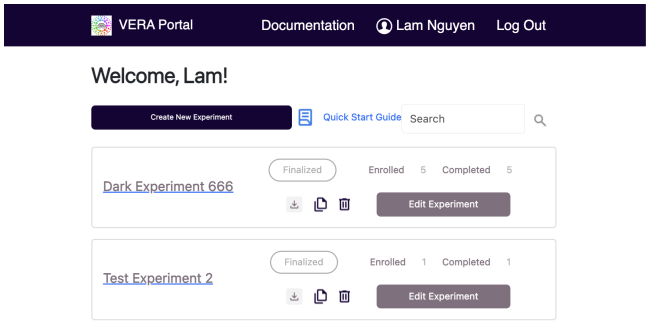

Figure 1: Application UI

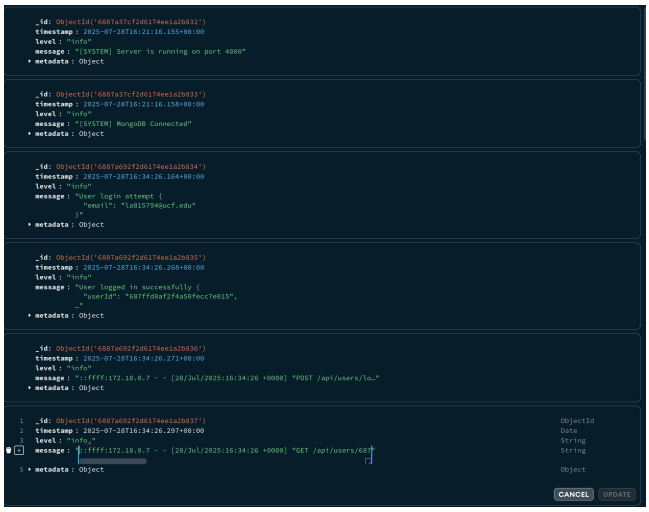

Figure 2: Backend Logs saved in a single location

Figure 3: Real-Time Logs in Terminal

In Figure 3 you can see the real-time logs that are generated in the terminal when the application is running. These logs provide information about the API requests and responses, as well as any errors that may occur. They are also used for debugging purposes during development.

The next steps will be to continue fine-tuning the integration of the LangChain framework and the Claude API to improve function, consistency and reliability. Once the application is able to consistently scan logs and identify anomalies, the next step will be to evaluate its performance using the evaluation metrics mentioned above.

6 Discussion and Challenges

7 Evaluation and Results

References

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